



Habib University
shaping futures

Instructions

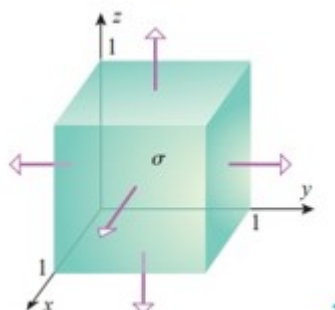
- The exam will begin at **5:45 PM** and conclude at **7:15 PM**, with a total duration of **1 hour and 30 minutes**. The exam is worth **30 points** in total.
- This is a closed books/notes exam. Any notes, books, bags should be kept at the front of the classroom before the start of the test.
- Use of cellphones, laptops or any other communicating device is prohibited. Your cellphones should be switched off and submitted to the invigilator, or kept at the front of the classroom before the start of the test.
- You are to attempt all question on the answer sheet provided. Your solutions should be comprehensible. Explain what you are doing, and if you use any results that you have studied in this course or Calculus 1 and 2, mention them explicitly.
- Use of calculators is allowed for this exam. Any form of communication/sharing among your peers is not allowed during the exam.
- If you want to communicate with the invigilator, raise your hand.
- You are not allowed to leave your desk without permission of the invigilator during the exam.
- Failure to abide by the instructions above would lead to immediate cancellation of the exam.

Student Name: _____

Instructor's Name: _____

Section: _____

1. (6 points) Suppose the surface σ of the unit cube in the figure below has an outward orientation.



In each part, determine:

- the outward unit normal vector on the specified face.
- whether the flux of the vector field $\vec{F} = z\hat{j}$ across that face is positive, negative, or zero.

Face	Outward Unit Normal Vector	Flux (Positive, Negative, or Zero)
$x = 1$		
$x = 0$		
$y = 1$		
$y = 0$		
$z = 1$		
$z = 0$		

2. (7 points) Sketch the following surface along with finding a parametric equation $\mathbf{r}(u, v)$ and a normal vector:

The ‘Rugby ball’ formed by rotating the curve $x = \cos z$, $y = 0$, $-\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$ around the z -axis.

3. (7 points) The area of the smooth surface S represented by:

$$\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$$

with

$$a \leq u \leq b, \quad c \leq v \leq d$$

is given by the surface integral:

$$\iint_S dA = \int_c^d \int_a^b |\vec{r}_u \times \vec{r}_v| \, du \, dv.$$

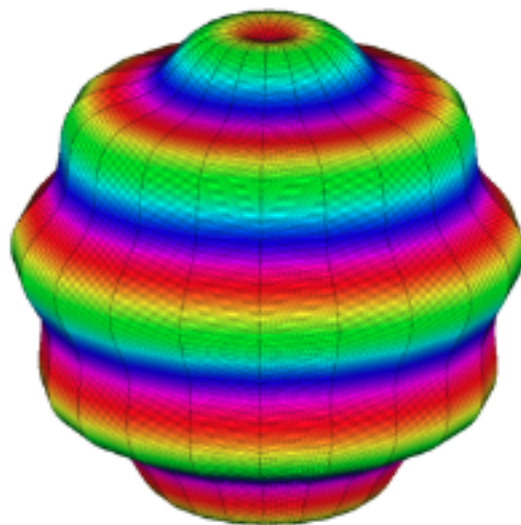
Use the formula above to find the surface area where the surface is the boundary of the closed hemisphere given by

$$x^2 + y^2 + z^2 \leq r^2, \quad z \geq 0.$$

4. (5 points) Let $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$. Show that the clockwise circulation of the field $\vec{F} = \vec{\nabla} f$ around the circle $x^2 + y^2 = a^2$ in the xy -plane is zero.
5. (5 points) Let k be a constant. Calculate the flux of the vector field:

$$\vec{F} = \vec{F}(x, y, z) = \left(2x + \tan^2 \left(\frac{\pi}{k} (y + z)^2 \right) \right) \hat{i} + (\ln |\sin x + \cos z| + 5y) \hat{j} + z \hat{k}$$

through the following closed surface, $\vec{r} = \vec{r}(\theta, \phi)$.



Assume the volume of the region to be V .